

Global Medical Device Supply Chain Analytics

Problem & Solution

The following business questions guided the design of the dashboard.

Procurement

Business Questions

- How much did we spend this year?
- Which supplier received the highest spend?
- Which plant purchases the most?
- Which buyer manages the most purchase orders?
- What is the monthly procurement trend?

Dashboard Solution

These questions are answered through:

- Executive KPI Cards
- Monthly Procurement Trend
- Spend by Supplier Analysis
- Spend by Plant Visualization
- Purchase Order Summary
- Procurement Analytics Dashboard

Supplier Performance

Business Questions

- Which supplier delivers late?
- Which supplier has the lowest Fill Rate?
- What is the average delivery delay?
- How do suppliers rank overall?

Dashboard Solution

These questions are answered using:

- Supplier Performance Score
- Fill Rate KPI
- On-Time Delivery %
- Average Delivery Delay
- Supplier Ranking Matrix
- Supplier Performance Scatter Plot

Inventory Management

Business Questions

- What is the current inventory value?
- Which materials are below safety stock?
- Which materials are overstocked?
- How is inventory distributed across warehouses?

Dashboard Solution

The dashboard provides:

- Inventory Value KPI
- Stock Status Classification
- Inventory by Warehouse
- Materials Below Safety Stock
- Inventory Distribution Charts

Demand Planning

Business Questions

- What is the forecast accuracy?
- Which materials have the largest forecast errors?
- What is the monthly demand trend?

Dashboard Solution

Demand planning is monitored through:

- Forecast vs Actual Comparison
- Monthly Demand Trend
- Forecast Accuracy KPI
- Material-Level Demand Analysis

Logistics

Business Questions

- What is the freight cost by carrier?
- What is the freight cost by transport mode?
- What is the average freight cost per shipment?

Dashboard Solution

Transportation analytics includes:

- Freight Cost by Carrier
- Freight Cost by Transport Mode
- Average Freight Cost
- Shipment Distribution
- Logistics KPI Cards

Overall Solution

The final Power BI solution integrates procurement, supplier, inventory, demand planning, and transportation data into a unified reporting platform. Interactive dashboards, DAX measures, and a star schema data model provide business users with timely, reliable, and actionable insights to support operational and strategic decision-making.

Key Outcomes

The implemented solution enables stakeholders to:

- Monitor procurement spending in real time.
- Evaluate supplier performance using standardized KPIs.
- Identify inventory risks before stock shortages occur.
- Compare forecasted demand against actual demand.
- Optimize transportation costs and carrier performance.
- Support executive decision-making through interactive dashboards.

Disclaimer: *This project was developed as a personal portfolio project using a simulated supply chain dataset. The company context, business scenarios, and analysis are intended to demonstrate Business Intelligence, data modeling, and dashboard development skills and do not represent actual operational data.*